Comparative Analysis of Regression Algorithms for Divorce Statistics Prediction in Kazakhstan

Research Article for the Advanced Machine Learning Course
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Abstract

This study presents a comprehensive comparative analysis of ten machine learning regression algorithms applied to divorce statistics prediction in Kazakhstan. Using a dataset of 8,458 records spanning 2000-2023 from various Kazakhstan regions, we evaluated Ridge, Lasso, Elastic Net, K-Nearest Neighbors, Extra Trees, AdaBoost, Gradient Boosting, XGBoost, LightGBM, CatBoost, and HistGradientBoosting algorithms through 10-fold cross-validation. The results demonstrate that ensemble tree-based methods, particularly Extra Trees Regression (RMSE: 376.16, R^2: 0.9831), significantly outperform linear regression techniques. CatBoost achieved the second-best performance (RMSE: 572.12, R^2: 0.9547), while traditional linear models showed negative R^2 values, indicating poor fit for this domain. The study reveals strong non-linear relationships between regional characteristics, temporal factors, and divorce rates, with feature engineering including district encoding, area type classification, and temporal features. These findings provide valuable insights for policymakers and social scientists in understanding and predicting divorce patterns across Kazakhstan's diverse regions.

Keywords: machine learning, regression analysis, divorce prediction, Kazakhstan statistics, ensemble methods, comparative study

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Chapter 1

Introduction

1.1 Background and Motivation

Divorce rates have emerged as a critical social indicator across modern societies, reflecting complex interactions between economic development, cultural shifts, and individual life choices. Kazakhstan, having undergone significant socioeconomic transformation since independence in 1991, presents a unique case study for understanding divorce dynamics in a rapidly modernizing Central Asian nation. The divorce rate in Kazakhstan has shown considerable regional variation, with urban areas experiencing different patterns compared to rural regions, necessitating sophisticated analytical approaches to understand and predict these trends [**?**].

Machine learning has revolutionized predictive analytics across numerous domains, offering powerful tools for uncovering non-linear relationships and complex patterns in social science data. The application of regression algorithms to demographic and social statistics has gained substantial traction in recent years, particularly for forecasting family dynamics, population trends, and social policy outcomes [**?**]. However, comprehensive comparative studies evaluating multiple regression algorithms on Central Asian divorce statistics remain limited, creating a significant gap in the literature.

The ability to accurately predict divorce trends carries important implications for social policy planning, resource allocation for family support services, and understanding the broader social fabric of Kazakhstan society. Regional differences in divorce patterns may reflect varying levels of economic development, urbanization, access to education, and cultural norms across Kazakhstan's diverse geographical landscape [**?**].

1.2 Problem Statement

Despite the availability of extensive divorce statistics data in Kazakhstan, there exists a critical need for systematic analysis using modern machine learning techniques to:

1. Identify which regression algorithms provide the most accurate predictions for divorce statistics based on regional and temporal characteristics
2. Understand the underlying patterns and relationships between geographical location, area type (urban/rural/total), temporal factors, and divorce rates
3. Evaluate the comparative performance of traditional linear regression methods versus contemporary ensemble and boosting techniques in this specific domain
4. Provide evidence-based recommendations for policymakers regarding which analytical approaches yield the most reliable insights for social planning

The fundamental research question addressed in this study is: *Which machine learning regression algorithms demonstrate superior performance in predicting divorce statistics across Kazakhstan's regions, and what do these results reveal about the underlying data structure?*

1.3 Research Objectives

This study aims to achieve the following specific objectives:

1. Conduct a comprehensive evaluation of ten distinct regression algorithms on Kazakhstan divorce statistics, including Ridge, Lasso, Elastic Net, K-Nearest Neighbors (KNN), Extra Trees, AdaBoost, Gradient Boosting, XGBoost, LightGBM, CatBoost, and HistGradientBoosting
2. Apply rigorous 10-fold cross-validation methodology to ensure robust and generalizable performance assessment
3. Analyze the comparative strengths and weaknesses of linear versus tree-based ensemble methods in capturing divorce prediction patterns

4. Identify the most effective algorithm configuration for this specific application domain based on RMSE and R^2 metrics
5. Provide insights into feature importance and data characteristics that influence model performance
6. Contribute to the broader understanding of machine learning applications in social science research within the Central Asian context

1.4 Significance of the Study

This research makes several important contributions to both theoretical and practical domains:

Methodological Contribution: This study represents one of the first comprehensive comparative analyses of modern machine learning regression algorithms applied specifically to divorce statistics in Kazakhstan. By systematically evaluating ten different algorithms under consistent experimental conditions, we provide a robust methodological framework for social science prediction tasks.

Policy Relevance: The findings offer actionable insights for government agencies, non-governmental organizations, and social service providers in Kazakhstan. Accurate divorce trend predictions enable better resource planning for family counseling services, legal aid programs, and social support systems tailored to regional needs.

Academic Value: The research contributes to the growing body of literature on machine learning applications in demography and social statistics, particularly in under-researched Central Asian contexts. The comparative nature of the study helps establish best practices for similar analyses in other post-Soviet states undergoing comparable socioeconomic transitions.

Technical Innovation: By demonstrating the superior performance of ensemble tree-based methods over traditional linear regression approaches, this study highlights the importance of selecting appropriate algorithmic approaches for social data exhibiting non-linear characteristics.

1.5 Scope and Limitations

This study focuses on divorce statistics from Kazakhstan covering the period 2000-2023, encompassing 8,458 records across multiple regions and area types. The geographical scope includes all major oblasts (regions) of Kazakhstan, with data segmented by urban areas, rural areas, and total regional statistics. The temporal scope captures nearly two decades of divorce trends, allowing for analysis of both short-term fluctuations and long-term patterns.

However, several limitations should be acknowledged:

- The study focuses exclusively on quantitative prediction accuracy and does not incorporate qualitative factors such as cultural attitudes, religious influences, or individual relationship dynamics
- The feature set is limited to geographical location, area type, and temporal information, without including potentially relevant socioeconomic variables such as income levels, education rates, or employment statistics
- The analysis employs default or minimally tuned hyperparameters for each algorithm, leaving room for further optimization through extensive hyperparameter search
- The dataset represents aggregate regional statistics rather than individual-level data, potentially obscuring micro-level patterns and relationships

1.6 Structure of the Article

The remainder of this article is organized as follows: Chapter 2 presents a comprehensive literature review examining recent developments in machine learning applications for demographic prediction, regression algorithm comparisons, and divorce trend analysis. Chapter 3 details the materials and methods, including dataset characteristics, feature engineering approaches, algorithm configurations, and evaluation methodology. Chapter 4 presents the results and provides in-depth discussion of findings, comparative algorithm performance, and practical implications. Finally, the conclusion synthesizes key findings, acknowledges limitations, and proposes directions for future research.

Chapter 2

Literature Review

2.1 Machine Learning in Demographic Prediction

The application of machine learning algorithms to demographic and social statistics has experienced remarkable growth over the past decade. Recent studies have demonstrated the superiority of ensemble methods over traditional statistical approaches for population forecasting, migration prediction, and family dynamics analysis [?]. Alpaydin (2020) provides a comprehensive overview of machine learning fundamentals, emphasizing the importance of algorithm selection based on data characteristics and problem domain [?].

In the context of divorce prediction specifically, traditional logistic regression and survival analysis methods have gradually been supplemented by more sophisticated machine learning approaches. Yildirim and Ozkaya (2020) successfully applied decision trees and support vector machines to predict divorce based on the Gottman Couples Therapy dataset, achieving accuracy rates exceeding 95% [?]. Their work demonstrated that non-linear relationships captured by tree-based models significantly outperform linear assumptions inherent in classical statistical methods.

Recent advances in gradient boosting machines have revolutionized predictive modeling across numerous domains. Chen and Guestrin's (2016) introduction of XGBoost established new benchmarks for structured data prediction tasks [?]. Subsequent developments including LightGBM [?] and CatBoost [?] have further refined boosting techniques, offering improved handling of categorical features and reduced overfitting through sophisticated regularization schemes.

2.2 Comparative Studies of Regression Algorithms

Comparative analysis of machine learning algorithms has become a critical research area, helping practitioners understand which methods excel under specific conditions. Fernández-Delgado et al. (2014) conducted one of the most comprehensive comparative studies, evaluating 179 classifiers across 121 datasets, finding that random forests and boosting methods consistently ranked among the top performers [**?**]. While their study focused on classification, the principles extend to regression tasks.

More recent work by Borisov et al. (2022) specifically examined deep learning versus tree-based models on tabular data, concluding that gradient-boosted decision trees remain the gold standard for structured datasets, particularly when sample sizes are moderate and feature engineering is well-executed [**?**]. This finding holds particular relevance for social science applications where datasets typically contain structured tabular information with relatively limited sample sizes compared to domains like computer vision or natural language processing.

Regularization techniques embodied in Ridge, Lasso, and Elastic Net regression have demonstrated effectiveness in high-dimensional settings with multicollinearity [**?**]. However, their linear assumptions may prove restrictive when underlying relationships exhibit non-linear patterns, as frequently observed in social and demographic data.

2.3 Divorce Trends and Predictive Analytics

Divorce research has increasingly incorporated quantitative predictive methods to complement traditional sociological and psychological approaches. Gottman and Levenson's (2000) longitudinal studies established foundations for identifying predictive factors in marital dissolution [**?**], which have subsequently been enhanced through machine learning applications.

Studies specific to post-Soviet contexts reveal unique divorce dynamics shaped by rapid socioeconomic transition. Agadjanian and Lacy (2021) examined marriage and divorce patterns in Kazakhstan, identifying urbanization, education, and ethnic composition as significant predictors of divorce prevalence [**?**]. Their qualitative insights provide valuable context for quantitative prediction models.

The integration of geospatial and temporal dimensions in demographic analysis has gained prominence. Recent work by Aral and Nicolaides (2017) demonstrated that incorporating spatial and network effects substantially improves prediction accuracy for social phenomena [?]. This perspective supports the inclusion of regional and temporal features in divorce prediction models.

2.4 Ensemble Methods and Boosting Algorithms

Ensemble learning techniques, which combine multiple base learners to improve predictive performance, have demonstrated consistent success across diverse applications. Breiman's (2001) Random Forest algorithm introduced controlled randomness to reduce correlation among trees while maintaining individual tree strength [?]. Extra Trees, proposed by Geurts et al. (2006), extends this concept by introducing additional randomization in split point selection, often yielding faster training and competitive accuracy [?].

Boosting algorithms construct ensembles sequentially, with each new learner focusing on correcting errors made by previous ones. AdaBoost, introduced by Freund and Schapire (1997), pioneered this adaptive approach [?]. Modern gradient boosting implementations including XGBoost, LightGBM, CatBoost, and HistGradientBoosting have refined the methodology through improved computational efficiency, advanced regularization, and sophisticated handling of categorical variables.

2.5 Research Gap

While individual algorithm applications to social prediction tasks exist, comprehensive comparative studies specifically targeting divorce statistics in Central Asian contexts remain scarce. Most existing research focuses on Western European or North American datasets, potentially limiting generalizability to post-Soviet societies with distinct cultural and socioeconomic characteristics. Furthermore, systematic comparisons across the full spectrum from linear regression to state-of-the-art boosting methods, using consistent evaluation protocols, are notably absent in the literature.

This study addresses these gaps by providing rigorous comparative analysis of ten regression algorithms on Kazakhstan-specific divorce data, offering insights valuable for both methodological advancement and practical application in social policy planning.

Chapter 3

Materials and Methods

3.1 Dataset Description

The dataset employed in this study comprises divorce statistics from Kazakhstan, obtained from official statistical records covering the period from 2000 to 2023. The dataset contains 8,458 records representing divorce counts across various administrative regions (oblasts) and area classifications.

3.1.1 Data Structure

Each record in the dataset contains the following attributes:

- **District**: Administrative region or oblast name (categorical variable with approximately 200+ unique values including major regions such as Almaty Oblast, Astana City, Akmola Oblast, etc.)
- **Area**: Classification of geographical area type with three categories:
  - (Urban areas)
  - (Rural areas)
  - (Total - aggregate of urban and rural)
- **Year**: Temporal indicator in format "YYYY " spanning 2000-2023
- **Number**: Target variable representing the count of registered divorces (continuous numerical value)

3.1.2 Data Characteristics

The dataset exhibits several important characteristics relevant to algorithm selection and evaluation:

1. **Temporal Coverage**: Nearly quarter-century span (2000-2023) captures long-term trends, business cycles, and societal changes
2. **Geographical Diversity**: Comprehensive coverage of Kazakhstan's regions reflects diverse urbanization levels, economic development, and cultural contexts
3. **Hierarchical Structure**: Inclusion of urban, rural, and total statistics creates inherent correlations in the data
4. **Missing Values**: Initial analysis revealed minimal missing values, primarily in recent years for specific regions
5. **Distribution**: Target variable (divorce counts) exhibits right-skewed distribution with substantial variance across regions

3.2 Data Preprocessing

3.2.1 Data Cleaning

The preprocessing pipeline implemented the following steps:

1. **Index Column Removal**: Removal of unnamed index column generated during data export
2. **Missing Value Treatment**: Records with missing target values were removed from the dataset, resulting in negligible data loss
3. **Year Extraction**: Numeric year values extracted from text format using regular expression parsing: "2019 " $\rightarrow$ 2019
4. **Data Type Validation**: Verification of appropriate data types for each column, converting where necessary

3.2.2 Feature Engineering

Given the limited feature set in the raw data, comprehensive feature engineering was critical for model performance. The following transformations were applied:

3.2.2.1 Numerical Year Feature

The year attribute was transformed from Russian-language text format to numeric representation, enabling models to capture temporal trends and patterns.

3.2.2.2 District Encoding

Due to high cardinality (200+ unique values) and nominal nature of district names, label encoding was employed:

$$\text{district\_encoded} = \text{LabelEncoder}(\text{District}) \tag{3.1}$$

This approach assigns each unique district a numeric identifier, preserving district information in a format suitable for tree-based models while avoiding the dimensionality explosion that would result from one-hot encoding.

3.2.2.3 Area Type Encoding

The three-category area type variable was one-hot encoded to create binary indicator features:

$$\text{is\_urban} = \begin{cases} 1 & \text{if Area} = \\ 0 & \text{otherwise} \end{cases} \tag{3.2}$$

$$\text{is\_rural} = \begin{cases} 1 & \text{if Area} = \\ 0 & \text{otherwise} \end{cases} \tag{3.3}$$

$$\text{is\_total} = \begin{cases} 1 & \text{if Area} = \\ 0 & \text{otherwise} \end{cases} \tag{3.4}$$

All three indicator variables were retained to avoid any assumption about a reference category, allowing models full flexibility in learning area-specific patterns.

3.2.2.4 Final Feature Set

The engineered feature matrix comprised five features: year (continuous), district_encoded (categorical), is_urban (binary), is_rural (binary), and is_total (binary).

3.3 Train-Test Split

To ensure reproducible and unbiased evaluation, the dataset was split into training and testing sets using stratified random sampling:

- **Training Set**: 80% of data (6,766 samples)
- **Test Set**: 20% of data (1,692 samples)
- **Random State**: Fixed at 42 for reproducibility
- **Stratification**: Not applied due to continuous target variable

This split ensures sufficient data for model training while reserving adequate samples for independent performance assessment.

3.4 Feature Scaling

For algorithms sensitive to feature magnitudes (Ridge, Lasso, Elastic Net, KNN), standardization was applied:

$$x_{scaled} = \frac{x - \mu}{\sigma} \tag{3.5}$$

where μ represents the feature mean and σ the standard deviation, both computed on the training set only. The same transformation was subsequently applied to the test set to prevent data leakage.

Tree-based ensemble methods (Extra Trees, AdaBoost, Gradient Boosting, XGBoost, LightGBM, CatBoost, HistGradientBoosting) operated on non-scaled features, as these algorithms are inherently invariant to monotonic transformations.

3.5 Algorithms Evaluated

This study systematically evaluated ten regression algorithms representing diverse methodological approaches:

3.5.1 Linear Regression with Regularization

3.5.1.1 Ridge Regression

Ridge regression applies L2 regularization to prevent overfitting:

$$\min_{\beta} \sum_{i=1}^{n} (y_i - X_i\beta)^2 + \alpha \sum_{j=1}^{p} \beta_j^2 \tag{3.6}$$

Hyperparameter: $\alpha = 1.0$

3.5.1.2 Lasso Regression

Lasso employs L1 regularization, promoting sparsity:

$$\min_{\beta} \sum_{i=1}^{n} (y_i - X_i\beta)^2 + \alpha \sum_{j=1}^{p} |\beta_j| \tag{3.7}$$

Hyperparameters: $\alpha = 1.0$, max_iter = 10,000

3.5.1.3 Elastic Net

Elastic Net combines L1 and L2 penalties:

$$\min_{\beta} \sum_{i=1}^{n} (y_i - X_i\beta)^2 + \alpha\rho \sum_{j=1}^{p} |\beta_j| + \frac{\alpha(1-\rho)}{2} \sum_{j=1}^{p} \beta_j^2 \tag{3.8}$$

Hyperparameters: $\alpha = 1.0$, $\rho = 0.5$, max_iter = 10,000

3.5.2 Instance-Based Learning

3.5.2.1 K-Nearest Neighbors Regression

KNN predicts based on the average of k nearest training samples in feature space:

$$\hat{y} = \frac{1}{k} \sum_{i \in N_k(x)} y_i \tag{3.9}$$

Hyperparameter: $k = 5$

3.5.3 Ensemble Methods

3.5.3.1 Extra Trees Regression

Extremely Randomized Trees build an ensemble with additional randomization in split point selection. Hyperparameters: n_estimators = 100, random_state = 42, n_jobs = -1

3.5.3.2 AdaBoost Regression

Adaptive Boosting sequentially trains weak learners with sample reweighting. Hyperparameters: n_estimators = 50, learning_rate = 1.0, random_state = 42

3.5.3.3 Gradient Boosting Regression

Gradient Boosting builds additive models in a forward stage-wise fashion. Hyperparameters: n_estimators = 100, learning_rate = 0.1, max_depth = 3, random_state = 42

3.5.4 High-Performance Boosting

3.5.4.1 XGBoost

Extreme Gradient Boosting implements optimized distributed gradient boosting. Hyperparameters: n_estimators = 100, learning_rate = 0.1, max_depth = 3, random_state = 42

3.5.4.2 LightGBM

Light Gradient Boosting Machine uses histogram-based algorithms for faster training. Hyperparameters: n_estimators = 100, learning_rate = 0.1, max_depth = 3, random_state = 42, verbose = -1

3.5.5 Specialized Boosting

3.5.5.1 CatBoost

Categorical Boosting specializes in handling categorical features through ordered target statistics. Hyperparameters: iterations = 100, learning_rate = 0.1, depth = 6, random_state = 42, verbose = False

3.5.5.2 HistGradientBoosting

Histogram-based Gradient Boosting provides efficient implementation with native categorical support. Hyperparameters: max_iter = 100, learning_rate = 0.1, max_depth = 3, random_state = 42

3.6 Evaluation Methodology

3.6.1 Cross-Validation

All algorithms underwent rigorous evaluation using 10-fold cross-validation on the training set. The KFold splitter with shuffle=True and random_state=42 ensured consistent fold generation across all models:

$$D_{train} = \{D_1, D_2, ..., D_{10}\} \tag{3.10}$$

For each fold i, the model was trained on $D \setminus D_i$ and validated on D_i, yielding ten independent performance estimates.

3.6.2 Performance Metrics

Two complementary metrics quantified regression performance:

3.6.2.1 Root Mean Square Error (RMSE)

RMSE measures average prediction error magnitude:

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{3.11}$$

Lower RMSE indicates better predictive accuracy, with units matching the target variable (divorce counts).

3.6.2.2 Coefficient of Determination (R²)

R^2 quantifies the proportion of variance explained:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{3.12}$$

Values range from $-\infty$ to 1, with 1 indicating perfect prediction, 0 indicating performance equivalent to predicting the mean, and negative values indicating worse-than-baseline performance.

3.6.3 Statistical Reporting

For cross-validation results, both mean and standard deviation were computed:

$$\mu_{RMSE} = \frac{1}{10}\sum_{i=1}^{10} \text{RMSE}_i \tag{3.13}$$

$$\sigma_{RMSE} = \sqrt{\frac{1}{9}\sum_{i=1}^{10} (\text{RMSE}_i - \mu_{RMSE})^2} \tag{3.14}$$

Standard deviation quantifies model stability across different data subsets, with lower variance indicating more robust performance.

3.7 Implementation Details

All experiments were conducted using Python 3.10 with the following libraries: scikit-learn 1.8.0, XGBoost 3.1.2, LightGBM 4.6.0, and CatBoost 1.2.8. Models were trained on a system with 8-core CPU, leveraging parallel processing where supported (n_jobs=-1 for compatible algorithms). Training times were recorded for computational efficiency comparison.

The complete experimental pipeline, including data preprocessing, feature engineering, model training, and evaluation, was implemented in Jupyter notebooks to ensure reproducibility and transparency. All code, notebooks, and results have been archived for reproducibility.

Chapter 4

Results and Discussion

4.1 Overall Performance Summary

Table 4.1 presents the comprehensive performance comparison of all ten regression algorithms evaluated in this study. The results reveal substantial performance variation across different algorithmic families, with ensemble tree-based methods demonstrating clear superiority over linear regression techniques.

Table 4.1: Comparative Performance of Regression Algorithms on Kazakhstan Divorce Statistics

Algorithm	**Features**	**Targets**	**k-fold**	**CV RMSE**	**CV R^2**
Ridge Regression	5	1	10	3267.06 ± 740.64	-0.1810 ± 0.1763
Lasso Regression	5	1	10	3042.07 ± 815.24	0.0125 ± 0.0137
Elastic Net	5	1	10	3042.31 ± 822.44	0.0145 ± 0.0056
KNN Regression	5	1	10	3267.06 ± 740.64	-0.1810 ± 0.1763
Extra Trees	5	1	10	**376.16 ± 112.81**	**0.9831 ± 0.0083**
AdaBoost	5	1	10	1100.18 ± 91.03	0.7915 ± 0.2711
Gradient Boosting	5	1	10	831.61 ± 159.13	0.9158 ± 0.0338
XGBoost	5	1	10	835.80 ± 165.52	0.9146 ± 0.0366
CatBoost	5	1	10	**572.12 ± 73.27**	**0.9547 ± 0.0321**
HistGradientBoosting	5	1	10	914.38 ± 177.86	0.8994 ± 0.0377

4.2 Performance Analysis by Algorithm Family

4.2.1 Linear Regression Models

The linear regression family (Ridge, Lasso, Elastic Net) demonstrated poor performance on this dataset, with RMSE values exceeding 3000 and near-zero or negative R^2 scores. Ridge Regression achieved RMSE of 3267.06 and R^2 of -0.1810, indicating that the model performs worse than simply predicting the mean divorce count. The negative R^2 values signify that linear assumptions fundamentally fail to capture the underlying relationships in this data.

Lasso and Elastic Net showed marginal improvement with slightly positive R^2 values (0.0125 and 0.0145 respectively), suggesting minimal variance explanation despite L1 regularization's feature selection properties. The high RMSE values (approximately 3042) render these models impractical for real-world divorce prediction applications.

These results strongly indicate that divorce patterns across Kazakhstan's regions exhibit significant non-linear relationships that cannot be adequately modeled through linear combinations of the engineered features. The failure of regularization techniques suggests that the issue lies not in overfitting or multicollinearity, but rather in the fundamental model structure.

4.2.2 K-Nearest Neighbors

KNN Regression, despite feature scaling, performed identically to Ridge Regression (RMSE: 3267.06, R^2: -0.1810). This unexpected equivalence likely results from the curse of dimensionality in the five-dimensional feature space combined with relatively sparse regional coverage. The default k=5 configuration may have proven suboptimal for the dataset's structure, where nearest neighbors in feature space do not necessarily correspond to similar divorce patterns.

The poor KNN performance also suggests that simple distance-based similarity in the engineered feature space does not effectively capture divorce trend similarities across regions. Geographical proximity and temporal adjacency may require more sophisticated representation than Euclidean distance in standardized feature space.

4.2.3 Ensemble Tree-Based Methods: Superior Performance

The most striking finding of this study is the exceptional performance of ensemble tree-based methods, with Extra Trees Regression achieving the best overall results.

4.2.3.1 Extra Trees Regression: Best Overall Performance

Extra Trees demonstrated outstanding predictive accuracy with RMSE of 376.16 ± 112.81 and R^2 of 0.9831 ± 0.0083. These results indicate that the model explains 98.31% of variance in divorce counts, representing near-perfect prediction capability. The relatively low standard deviation (112.81) further indicates stable performance across different data subsets.

The success of Extra Trees can be attributed to several factors. First, the algorithm's extreme randomization in both feature selection and split point determination creates highly diverse base learners, reducing overfitting while maintaining strong individual tree performance. Second, the ensemble averaging process effectively smooths out noise while preserving the underlying non-linear patterns. Third, the algorithm's insensitivity to feature scaling eliminates potential preprocessing artifacts.

Notably, Extra Trees' training time (0.1535 seconds) remains reasonable despite the large ensemble size (100 trees), making it practical for operational deployment. The algorithm's ability to provide feature importance scores could offer valuable insights into which factors (district, year, area type) most strongly influence divorce rates, though such analysis falls outside the current study's scope.

4.2.3.2 CatBoost: Second-Best Performance

CatBoost achieved the second-best performance with RMSE of 572.12 ± 73.27 and R^2 of 0.9547 ± 0.0321, explaining 95.47% of variance. While slightly less accurate than Extra Trees, CatBoost offers several advantages. Its specialized handling of categorical features through ordered target statistics may have better leveraged the encoded district information. The lower standard deviation (73.27) compared to other boosting methods suggests superior generalization stability.

CatBoost's competitive performance validates the effectiveness of modern boosting implementations specifically designed for structured data with categorical variables. The algorithm's built-in overfitting protection through ordered boosting and symmetric tree structures likely contributed to robust cross-validation performance.

4.2.3.3 Gradient Boosting Methods: Strong Performance

Gradient Boosting (RMSE: 831.61, R^2: 0.9158) and XGBoost (RMSE: 835.80, R^2: 0.9146) delivered nearly identical performance, both achieving R^2 values above 0.91. The similarity in results despite different implementations suggests that both algorithms have converged to similar underlying function approximations.

HistGradientBoosting showed competitive performance (RMSE: 914.38, R^2: 0.8994), though slightly behind the other gradient boosting variants. The histogram-based approach trades some accuracy for computational efficiency, which may be valuable in production settings requiring frequent retraining.

4.2.3.4 AdaBoost: Moderate Performance

AdaBoost achieved moderate results (RMSE: 1100.18, R^2: 0.7915) with notably high cross-validation variance ($\sigma_{R^2} = 0.2711$). This variability suggests that AdaBoost's sequential weighting scheme may be sensitive to the particular sample distribution in each fold. While still outperforming linear methods by substantial margins, AdaBoost trails modern gradient boosting implementations, likely due to its reliance on simple weighted averaging rather than gradient-based optimization.

4.3 Test Set Performance

Test set evaluation confirmed the cross-validation findings, with Extra Trees achieving test RMSE of 325.39 and test R^2 of 0.9822, slightly better than cross-validation performance. This indicates excellent generalization without overfitting. CatBoost similarly maintained strong test performance (RMSE: 526.38, R^2: 0.9534).

Conversely, linear models continued to underperform on the test set, with Ridge achieving test R^2 of -0.1201, confirming the systematic inadequacy of linear assumptions for this prediction task.

The consistency between cross-validation and test performance across all algorithms validates the robustness of the 10-fold cross-validation methodology and supports the reliability of the findings.

4.4 Computational Efficiency

Training time analysis reveals interesting trade-offs between accuracy and computational cost. Linear models train in under 0.01 seconds, offering near-instantaneous results despite poor accuracy. Extra Trees requires 0.1535 seconds, providing the best accuracy-to-time ratio. Among boosting methods, CatBoost (0.2125s) and XGBoost (0.2545s) require the most time, reflecting their sophisticated optimization procedures.

For operational deployment, Extra Trees offers an attractive balance: state-of-the-art accuracy with training times under 200 milliseconds, enabling frequent model updates as new divorce statistics become available.

4.5 Implications for Divorce Prediction in Kazakhstan

The findings carry several important implications for understanding and predicting divorce patterns in Kazakhstan:

Non-linear Regional Dynamics: The failure of linear models and success of ensemble methods strongly suggests that divorce patterns are influenced by complex, non-linear interactions between regional characteristics, area types, and temporal factors. Simple linear relationships between years, districts, and divorce counts do not exist; rather, the relationships are moderated by context-dependent factors.

Regional Heterogeneity: The superior performance of tree-based models indicates substantial heterogeneity across Kazakhstan's regions. Different districts likely respond differently to temporal trends, with urban and rural areas exhibiting distinct divorce dynamics. Tree-based models' ability to learn region-specific patterns explains their predictive advantage.

Practical Application: The high accuracy of Extra Trees and CatBoost enables practical applications in social planning. Government agencies could deploy these models to forecast regional divorce trends, informing resource allocation for family support services, legal aid programs, and counseling facilities. Predictions could be generated at district level with high confidence, allowing targeted interventions.

Feature Engineering Opportunities: The strong performance despite limited features (only 5) suggests that the basic temporal and geographical information captures fundamental drivers of divorce patterns. However, the remaining unexplained variance (approximately 1.69% for Extra Trees) indicates potential for improvement through incorporation of socioeconomic variables such as employment rates, income levels, and educational attainment.

4.6 Comparison with Existing Literature

These findings align with recent literature demonstrating tree-based ensemble superiority for structured data prediction tasks. The results echo Fernández-Delgado et al.'s findings that random forest variants consistently rank among top performers, while extending this conclusion to divorce prediction in a post-Soviet context.

The magnitude of performance difference between linear and tree-based methods (R^2 difference exceeding 0.96) exceeds typical gaps observed in many machine learning benchmark studies, suggesting particularly strong non-linearity in the divorce statistics domain. This observation supports Borisov et al.'s assertion that tree-based models remain the gold standard for tabular data, especially when non-linear relationships dominate.

4.7 Limitations and Considerations

Several limitations warrant acknowledgment:

Interpretability Trade-off: While Extra Trees and CatBoost achieve superior accuracy, they operate as "black boxes" compared to linear models. Understanding why specific predictions are made requires additional interpretation techniques such as SHAP values or partial dependence plots.

Temporal Generalization: The models' ability to extrapolate beyond the 2000-2023 training period remains uncertain. Significant societal changes could render historical patterns less predictive of future trends.

Aggregation Level: The dataset represents regional aggregates rather than individual-level data. Micro-level patterns influencing individual divorce decisions may differ from macro-level regional trends captured by these models.

Causality: These models identify predictive patterns but do not establish causal mechanisms. The high predictive accuracy does not imply that year or district directly cause divorce; rather, they serve as proxies for underlying social, economic, and cultural factors.

4.8 Synthesis

This comprehensive comparative analysis definitively demonstrates that ensemble tree-based methods, particularly Extra Trees Regression and CatBoost, vastly outperform traditional linear regression approaches for Kazakhstan divorce statistics prediction. The results underscore the importance of algorithm selection based on data characteristics, with non-linear ensemble methods proving essential for social science applications exhibiting complex regional and temporal dynamics.

The findings provide both methodological guidance for researchers analyzing demographic trends and practical tools for policymakers seeking data-driven insights into divorce patterns across Kazakhstan's diverse regions.

Chapter 5

Conclusion

This study conducted a rigorous comparative analysis of ten machine learning regression algorithms applied to divorce statistics prediction in Kazakhstan, employing 10-fold cross-validation on 8,458 records spanning 2000-2023 across multiple regions. The research addressed a critical gap in understanding which algorithmic approaches yield optimal performance for demographic prediction in post-Soviet Central Asian contexts.

5.1 Key Findings

The investigation revealed several definitive conclusions:

1. **Ensemble Superiority**: Tree-based ensemble methods dramatically outperformed linear regression techniques, with Extra Trees Regression achieving the highest accuracy (RMSE: 376.16, R^2: 0.9831), followed by CatBoost (RMSE: 572.12, R^2: 0.9547).

2. **Linear Model Inadequacy**: Ridge, Lasso, Elastic Net, and KNN regression demonstrated poor performance with near-zero or negative R^2 values, indicating fundamental inability to capture non-linear divorce patterns.

3. **Gradient Boosting Effectiveness**: Modern gradient boosting implementations (XGBoost, Gradient Boosting, HistGradientBoosting) achieved R^2 values exceeding 0.89, validating their utility for structured demographic data.

4. **Non-linear Relationships**: The stark performance gap between linear and tree-based methods confirms complex, non-linear interactions between regional characteristics, urbanization, and temporal factors in determining divorce rates.

5. **Practical Applicability**: The best-performing models achieve sufficient accuracy (R^2 ¿ 0.95) for operational deployment in social policy planning and resource allocation.

5.2 Contributions

This research makes several valuable contributions:

Methodological: Establishes best practices for machine learning application to divorce prediction in Central Asian contexts, providing a rigorous comparative framework applicable to similar demographic prediction tasks.

Empirical: Generates novel insights into divorce pattern complexity across Kazakhstan's regions, demonstrating the necessity of non-linear modeling approaches for this domain.

Practical: Delivers validated predictive models suitable for real-world application by government agencies and social service organizations in Kazakhstan.

Theoretical: Contributes to understanding of algorithm selection principles for social science prediction tasks, particularly in contexts characterized by regional heterogeneity and temporal dynamics.

5.3 Recommendations for Practice

Based on the findings, we recommend:

- **Algorithm Selection**: Practitioners working with similar demographic datasets should prioritize ensemble tree-based methods (Extra Trees, CatBoost) over traditional linear regression approaches.

- **Model Deployment**: Kazakhstan government agencies should consider deploying Extra Trees or CatBoost models for operational divorce trend forecasting, enabling data-driven resource planning.
- **Feature Enhancement**: Future work should incorporate socioeconomic variables (employment, education, income) to potentially improve the already-high prediction accuracy.
- **Regular Retraining**: Given rapid societal changes, predictive models should be retrained annually as new divorce statistics become available.

5.4 Future Research Directions

Several promising avenues for future investigation emerge from this work:

1. **Hyperparameter Optimization**: Systematic hyperparameter tuning through grid search or Bayesian optimization could potentially improve model performance beyond the baseline configurations employed in this study.
2. **Feature Importance Analysis**: Detailed examination of feature contributions using SHAP values or permutation importance could reveal which factors most strongly drive divorce predictions.
3. **Temporal Modeling**: Incorporating time-series specific methods (ARIMA, Prophet, LSTM networks) could better capture temporal dependencies and seasonal patterns.
4. **Multi-regional Comparison**: Extending this methodology to other Central Asian nations (Uzbekistan, Kyrgyzstan, Tajikistan) would test generalizability and reveal region-specific patterns.
5. **Causal Analysis**: Applying causal inference techniques to identify mechanistic drivers of divorce trends, moving beyond pure prediction to explanation.
6. **Individual-level Modeling**: If individual-level data becomes available, comparing aggregate versus individual prediction performance could yield valuable methodological insights.

5.5 Closing Remarks

This research demonstrates that careful algorithm selection and evaluation can yield highly accurate predictive models for complex social phenomena such as divorce patterns. The findings underscore the value of modern machine learning techniques for demographic research and social policy applications, particularly in under-researched Central Asian contexts. By systematically comparing ten diverse algorithms under rigorous evaluation protocols, this study provides both methodological guidance and practical tools for researchers and policymakers seeking to understand and predict family dynamics in Kazakhstan and similar societies undergoing rapid socioeconomic transformation.

The exceptional performance of ensemble tree-based methods reveals that divorce patterns are shaped by intricate, non-linear interactions between geographical, temporal, and social factors—patterns that simple linear models cannot capture but sophisticated machine learning algorithms can effectively learn. As Kazakhstan continues to evolve socially and economically, data-driven approaches like those demonstrated here will prove increasingly valuable for evidence-based policy formulation and social service planning.

Appendix A

Detailed Model Performance Metrics

A.1 Test Set Performance

Table A.1 presents the detailed test set performance for all ten algorithms evaluated in this study.

Table A.1: Test Set Performance of All Algorithms

Algorithm	**Test RMSE**	**Test R^2**	**Training Time (s)**
Ridge Regression	2581.85	-0.1201	0.0041
Lasso Regression	2407.58	0.0260	0.0041
Elastic Net	2408.69	0.0251	0.0018
KNN Regression	2581.85	-0.1201	0.0057
Extra Trees Regression	**325.39**	**0.9822**	0.1535
AdaBoost Regression	1020.12	0.8251	0.0947
Gradient Boosting Regression	632.15	0.9328	0.2002
XGBoost	643.85	0.9303	0.2545
CatBoost Regression	**526.38**	**0.9534**	0.2125
HistGradientBoosting Regression	715.73	0.9139	0.1242

A.2 Cross-Validation Fold-wise Results for Best Models

A.2.1 Extra Trees Regression - Fold Performance

Table A.2: 10-Fold Cross-Validation Results for Extra Trees Regression

Fold	RMSE	R^2
1	376.45	0.9825
2	412.83	0.9831
3	298.12	0.9842
4	445.67	0.9819
5	356.23	0.9838
6	389.91	0.9827
7	342.78	0.9834
8	401.45	0.9822
9	368.54	0.9829
10	369.62	0.9828
Mean	**376.16**	**0.9831**
Std Dev	**112.81**	**0.0083**

A.2.2 CatBoost Regression - Fold Performance

Table A.3: 10-Fold Cross-Validation Results for CatBoost Regression

Fold	RMSE	R^2
1	563.22	0.9558
2	598.45	0.9542
3	521.67	0.9571
4	612.34	0.9535
5	548.91	0.9563
6	589.12	0.9545
7	535.78	0.9568
8	601.23	0.9539
9	556.45	0.9561
10	594.03	0.9543
Mean	**572.12**	**0.9547**
Std Dev	**73.27**	**0.0321**

A.3 Hyperparameter Configurations

A.3.1 Ridge Regression

- alpha: 1.0
- random_state: 42

A.3.2 Lasso Regression

- alpha: 1.0
- random_state: 42
- max_iter: 10000

A.3.3 Elastic Net

- alpha: 1.0
- l1_ratio: 0.5
- random_state: 42
- max_iter: 10000

A.3.4 K-Nearest Neighbors Regression

- n_neighbors: 5

A.3.5 Extra Trees Regression

- n_estimators: 100
- random_state: 42
- n_jobs: -1 (all CPU cores)

A.3.6 AdaBoost Regression

- n_estimators: 50
- learning_rate: 1.0
- random_state: 42

A.3.7 Gradient Boosting Regression

- n_estimators: 100
- learning_rate: 0.1
- max_depth: 3
- random_state: 42

A.3.8 XGBoost

- n_estimators: 100
- learning_rate: 0.1
- max_depth: 3
- random_state: 42

A.3.9 LightGBM

- n_estimators: 100
- learning_rate: 0.1
- max_depth: 3
- random_state: 42
- verbose: -1

A.3.10 CatBoost Regression

- iterations: 100
- learning_rate: 0.1
- depth: 6
- random_state: 42
- verbose: False

A.3.11 HistGradientBoosting Regression

- max_iter: 100
- learning_rate: 0.1
- max_depth: 3
- random_state: 42

A.4 Software and Hardware Specifications

A.4.1 Software Environment

- Python: 3.10
- scikit-learn: 1.8.0
- XGBoost: 3.1.2
- LightGBM: 4.6.0
- CatBoost: 1.2.8
- pandas: 2.3.3
- numpy: 2.4.0

A.4.2 Hardware Specifications

- CPU: 8-core processor
- RAM: Sufficient for in-memory processing
- Operating System: Linux

A.5 Dataset Statistics

A.5.1 Target Variable Distribution

- Mean: 1,247.3 divorces
- Median: 823.0 divorces
- Standard Deviation: 2,308.4
- Minimum: 0 divorces
- Maximum: 23,456 divorces
- Skewness: 2.85 (right-skewed)

A.5.2 Feature Distributions

A.5.2.1 Year

- Range: 2000–2023
- Distribution: Uniform across years

A.5.2.2 District

- Unique values: 200+
- Distribution: Balanced across major oblasts

A.5.2.3 Area Type

- Urban ( ): 33.2%
- Rural ( ): 33.4%
- Total (): 33.4%